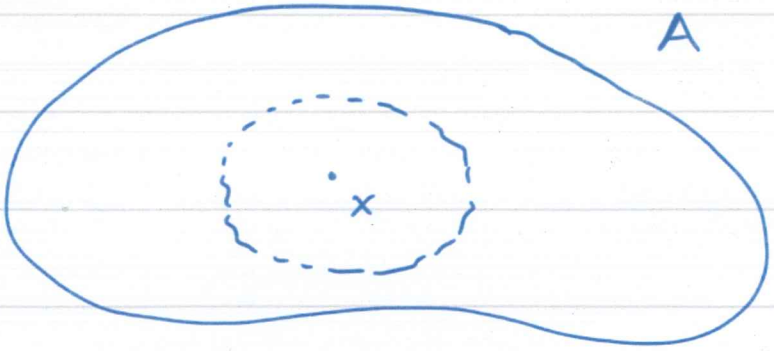
$y_n \not\to d$ ($\{ d \in \{d\}$ but $y_n \notin \{d\}$ for any $n \geq 1$)

Interior, exterior, boundary & closure of a set

Definition: Let (X, τ) be a topological space, A a subset of X & x be a point in X. If there exists an open set $U \in \tau$ satisfying $x \in U \subseteq A$ then x is said to be an interior point of A.

The set of all interior points of A is called the interior of A & it is denoted by $\text{int}(A)$ or $\mathring{A}$.



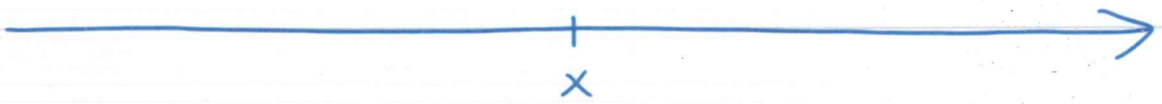
Note that $\mathring{A} \subseteq A$.

If A is a nonempty set, $\mathring{A}$ may or may not be empty:

Example: Let $X = \mathbb{R}$ & let τ_1 & τ_2 be the usual & discrete topologies on $X = \mathbb{R}$, respectively. Let $A = \{x\}$. Then find $\mathring{A}$ in (X, τ_1) & (X, τ_2).

Soln

$\mathring{A}$ in $(X, \tau_1) = (\mathbb{R}, \tau_1)$



Because we cannot find any open interval I with $x \in I \subseteq \{x\} = A$

$\mathring{A} = \emptyset$.

$\mathring{A}$ in $(\mathbb{R}, \tau_2)$

Since any subset of $\mathbb{R}$ is an elt.

of τ_2 $\quad \{x\} = A \in \tau_2$. Thus

$x \in A \subseteq A$.

$\Rightarrow \quad x \in \mathring{A} \quad \Rightarrow \quad \mathring{A} = A$.

Theorem: Let (X, τ) be a topological space & A a subset of X. Then

(i) $\mathring{A}$ is the union of all open subsets of A.

(ii) $\mathring{A}$ is open

(iii) $\mathring{A}$ is the largest open subset of A.

(iv) A is open $\Leftrightarrow A = \mathring{A}$

(v) $(\mathring{A})^{\circ} = \mathring{A}$

Proof:

(i) Let $\mathcal{A} = \{T \mid T \in \tau \ \& \ T \subseteq A\}$ & let $G = \bigcup \mathcal{A}$. We want to show that $\mathring{A} = G$:

Let $x \in \mathring{A}$. Then there exists an open set $U \in \mathcal{T}$ s.t. $x \in U \subseteq A$.

$\Rightarrow U \in \mathcal{A} \Rightarrow x \in \bigcup \mathcal{A} = G$

$\Rightarrow \mathring{A} \subseteq G$ (1)

Let $x \in G = \bigcup \mathcal{A} \Rightarrow$ There exists a $T \in \mathcal{T}$ with $T \subseteq A$ s.t. $x \in T$. i.e. $x \in T \subseteq A \Rightarrow x \in \mathring{A}$.

$\Rightarrow G \subseteq \mathring{A}$ (2)

By (1) & (2) we have $\mathring{A} = G$.

(ii) By (i) G is an open set, i.e. $G \in \mathcal{T}$. Moreover $G = \mathring{A}$. Thus $\mathring{A}$ is an open set.

(iii) It follows from (i).

(iv) $\Longleftarrow$ Let A be open. We know that $\mathring{A}$ is the largest open subset of A by (iii).

But A is open & $A \subseteq A$. Thus

$A \subseteq \mathring{A}$ (1)

$\mathring{A} \subseteq A$ (2)

By (1) & (2) $A = \mathring{A}$.

$\Leftarrow$ If $A = \mathring{A}$ because $\mathring{A}$ is open

by (ii) A is open.

(v) Since $\mathring{A}$ is open by (ii)

$(\mathring{A})^{\circ} = \mathring{A}$ by (iv).

Theorem: Let $(X, \mathcal{T})$ be a topological space & A & B be subsets of X.

Then we have;

(i) If $A \subseteq B$ then $\mathring{A} \subseteq \mathring{B}$

(ii) $(A \cap B)^{\circ} = \mathring{A} \cap \mathring{B}$

(iii) $\mathring{A} \cup \mathring{B} \subseteq (A \cup B)^{\circ}$.

Proof:

(i) Let $A \subseteq B$. Then

$\mathring{A} \subseteq A \subseteq B$ Moreover $\mathring{B}$ is the largest open subset of B.

$\Rightarrow \mathring{A} \subseteq \mathring{B}$.

(ii) $A \cap B \subseteq A, B$

$\Rightarrow (A \cap B)^{\circ} \subseteq \mathring{A}, \mathring{B}$ by (i)

$\Rightarrow (A \cap B)^{\circ} \subseteq \mathring{A} \cap \mathring{B}$ (1)

Because $\mathring{A} \subseteq A$, $\mathring{B} \subseteq B$,

$\mathring{A} \cap \mathring{B} \subseteq A \cap B$. By (i)

$(\mathring{A} \cap \mathring{B})^{\circ} = \mathring{A} \cap \mathring{B} \subseteq (A \cap B)^{\circ}$ (2)

By (1) & (2)

$(A \cap B)^{\circ} = \mathring{A} \cap \mathring{B}$.

(iii) Since $A, B \subseteq A \cup B$.

$\mathring{A}, \mathring{B} \subseteq (A \cup B)^{\circ}$

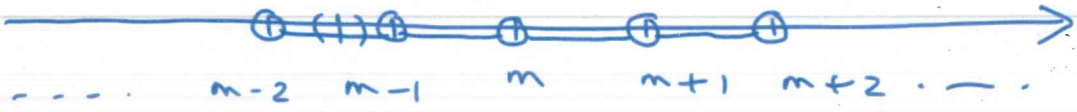
$$\Rightarrow \quad \mathring{A} \cup \mathring{B} \subseteq (A \cup B)^{\circ}.$$

Example: Let $(\mathbb{R}, \tau)$ be the usual topological space, $A = \mathbb{Z}$ & $B = \mathbb{R} \setminus \mathbb{Z}$. Find $\mathring{A} \cup \mathring{B}$ & $(A \cup B)^{\circ}$

Soln

$$\mathring{A} = \emptyset$$





$$\mathring{B} = (\mathbb{R} \setminus \mathbb{Z})^{\circ} = \mathbb{R} \setminus \mathbb{Z} = B.$$

$$\mathring{A} \cup \mathring{B} = B$$

$$(A \cup B)^{\circ} = \mathring{\mathbb{R}} = \mathbb{R}$$

$\therefore \quad \mathring{A} \cup \mathring{B} \neq (A \cup B)^{\circ}$.

Example: Let $(\mathbb{R}, \tau)$ be the usual topological space. $[a,b]^\circ = (a,b)$.

Hence $[a,b]$ is not an open set.

However $[a,b]$ is a closed set.

Because $\mathbb{R} \setminus [a,b] = \underbrace{(-\infty, a)}_{\text{open}} \cup \underbrace{(b, \infty)}_{\text{open}} \in \tau$.

Similarly $(-\infty, a]$ & $[a, \infty)$ are closed sets: $\mathbb{R} \setminus (-\infty, a] = (a, \infty) \in \tau$ & $\mathbb{R} \setminus [a, \infty) = (-\infty, a) \in \tau$.

Moreover $(-\infty, a]^\circ = (-\infty, a)$ & $[a, \infty)^\circ = (a, \infty)$.

Example: Let $X = \{a, b, c, d, e\}$, $\tau = \{\phi, \{a\}, \{a, b\}, \{a, c, d\}, \{a, b, c, d\}, \{a, b, e\}, X\}$. Consider $A = \{a, b, c\}$, $B = \{a, b, c, e\}$ & $C = \{a, c, e\}$.

Determine the interiors of A, B & C.

Sol^n

Recall that interior of a set is the largest open subset of that set.

$\mathring{A} = \{a, b\}$

$B^{\circ} = \{a, b, e\}$

$C^{\circ} = \{a\}$

OR

Use the defⁿ of interior of a set.

We know that the set of all interior points of a set is called the interior of that set.

$A = \{a, b, c\}$. Because $\mathring{A} \subseteq A$ we look at the pt's of A in order to find interior points of A:

Is $a \in \mathring{A}$? $a \in \{a\} \subseteq A$. Here $\{a\} \in \tau$. Thus $a \in \mathring{A}$.

Is $b \in \mathring{A}$? Since $\{a, b\} \in \tau$ & $b \in \{a, b\} \subseteq A$, $b \in \mathring{A}$.

Is $c \in \mathring{A}$? $\{a, c, d\}$, $\{a, b, c, d\}$, X are all possible τ-open sets containing c. But none of these sets is contained in A. Thus $c \notin \mathring{A}$.

$\therefore \mathring{A} = \{a, b\}$.

Definition: Let (X, τ) be a topological space & $A \subseteq X$. The interior points of $X \setminus A$ are called exterior points of A. The set of all exterior points of A is called the exterior of A & it is denoted by $ext(A)$. Hence

$$ext(A) = (X \setminus A)^{\circ}.$$

Clearly $\overset{\circ}{A} \cap ext(A) = \phi$ & $A \cap ext(A) = \phi$.

$$\left.\begin{array}{l} A \cap (X \setminus A) = \phi \\ (X \setminus A)^{\circ} \subseteq X \setminus A \end{array}\right\} A \cap (X \setminus A)^{\circ}$$

$$= A \cap ext(A)$$

$$= \phi.)$$

Definition: Let (X, τ) be a topological space & $A \subseteq X$. If

a point $x \in X$ is neither an interior pt. nor an exterior pt. of A then it is called a boundary point of A. The set of all boundary points of A is called the boundary of A & it is denoted by ∂A or $bdy(A)$.

So

$$\partial A = X \setminus [\mathring{A} \cup ext(A)].$$